

UNIVERSITY OF ENGINEERING AND TECHNOLOGY, LAHORE
Department of Mechanical Engineering

Industrial Internship Report

One-Month Internship at World Machinery Store
(Diesel Generators & Solar Power Solutions), Lahore

**Generator Load Testing, Performance Analysis, Routine
Maintenance, and Standalone Solar Integration**

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Institution	University of Engineering and Technology, Lahore
Organisation	World Machinery Store, Lahore
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Executive Summary

During August 2026 I completed a four-week industrial internship at **World Machinery Store (WMS)**, a Lahore-based supplier of new and reconditioned diesel generating sets and a provider of turnkey solar power systems. My placement covered three areas of work:

- **Generator load testing, performance analysis and routine maintenance.** I took part in the pre-delivery inspection (PDI) and resistive load-bank testing of three generating sets (55–110 kW) and carried out scheduled preventive-maintenance tasks. A full step-load test that I ran on a Cummins-powered 62.5 kVA set gave a minimum specific fuel consumption of 264 g/kWh and a peak brake-thermal efficiency of 32%, both between 75% and 100% load, with frequency held inside $\pm 1\%$ and line voltage inside $\pm 3\%$ across the full range.
- **Applied research** on “*Performance Analysis and Operational Optimization of Diesel Generator Sets under Variable Load Conditions*”, which I used to derive practical operating and maintenance recommendations for WMS and its customers.
- **Standalone solar integration and backup solutions.** I assisted with the DC-side termination and mechanical inspection of an 11.8 kWp three-phase hybrid array, modelled its annual yield for Lahore ($\approx 16,200$ kWh; specific yield 1,374 kWh/kWp), and prepared a techno-economic comparison showing that a solar-hybrid backup can cut the workshop’s standby-power running cost by about 80% and its operational CO₂ by about 83%, with a simple payback below three years.

Acknowledgement

I thank the management of World Machinery Store for the opportunity to train with the company, and in particular Mr. Salman Farooq (Technical Advisor) for his supervision, along with the workshop technicians and the solar installation team for their guidance and patience. I am also grateful to the Department of Mechanical Engineering, UET Lahore, for facilitating the industrial-training requirement.

1 Introduction

1.1 The organisation

World Machinery Store (WMS) is a Lahore-based trading and engineering-services company operating in the standby-power sector. Its activities cover:

- import, reconditioning and sale of diesel generating sets (typically 15–500 kVA) built around Cummins, Perkins and comparable engines;
- load-bank testing and pre-delivery inspection, installation, commissioning, and annual maintenance contracts (AMCs);
- turnkey solar power systems – on-grid, off-grid and hybrid – including design, supply, mounting, and net-metering support.

The workshop serves industrial, commercial and residential customers across Punjab, where frequent utility load-shedding makes reliable backup power a routine requirement.

1.2 Objectives of the internship

My objectives for the placement were:

- to understand how a diesel generating set is assembled, tested and maintained, and how acceptance testing is performed against recognised standards;
- to carry out and document a generator load test and to analyse the resulting performance data;
- to study how variable load affects generator efficiency, emissions and maintenance, and to translate that into operating recommendations;
- to learn the fundamentals of standalone/hybrid solar power systems and to evaluate solar as a backup-power option.

1.3 Scope of work and schedule

My internship ran for four weeks, on a six-day week, and was split between the generator workshop and the solar division. Table 1 summarises my weekly programme.

Table 1. Weekly programme of the internship.

Week	Focus	Main activities
Week 1 (01–06 Aug)	Orientation & familiarisation	HSE induction; study of genset components (engine, alternator, control panel, ATS); shadowed two PDIs; review of Cummins 4B/6B service literature.
Week 2 (08–13 Aug)	Load-bank testing	Instrumentation and test procedure; full step-load test on set G-01; data logging and analysis; assisted AVR/governor adjustment on set G-02.
Week 3 (15–21 Aug)	Preventive maintenance & diagnostics	Oil, fuel and air-filter service; cooling-system and battery checks; fault diagnosis on set G-03; start of the variable-load literature review.
Week 4 (22–30 Aug)	Solar division	DC string termination at a WOXEN hybrid inverter; module mounting and earthing inspection; energy-yield modelling; diesel-vs-hybrid backup study; report compilation.

2 Generator Load Testing and Performance Analysis

2.1 Test setup and method

Every set at WMS is load-tested before dispatch to confirm that it delivers its rated output and that all engine and alternator parameters stay within limits. I ran this test on set G-01 and assisted on two others, following the manufacturer's PDI checklist and the guidance of ISO 8528-1/-5 (generating-set ratings, performance classes and transient response).

- **Load bank:** portable 75 kVA resistive bank with 5 kW and 1 kW trim steps, plus a reactive module that I used to load the set at 0.8 power factor (lagging) so that the test represents a real installation.
- **Instrumentation:** three-phase power analyser (V, I, kW, kVA, pf, frequency), optical tachometer, infrared pyrometer for exhaust and coolant temperatures, engine oil-pressure gauge, and a graduated day-tank with a stopwatch for volumetric fuel measurement.
- **Procedure:** I started the set cold, ran a five-minute warm-up and let it stabilise at no load, then applied load in 25% steps, holding each for 15 minutes to reach thermal stability and averaging the readings over the final three minutes. I applied a 110% overload for 60 seconds per ISO 8528, then carried out a controlled cool-down and shutdown.
- **Ambient conditions during the G-01 test:** 34–39 °C, $\approx 55\%$ relative humidity, site elevation 217 m. I noted the standard derating factors for temperature and altitude but did not apply them to the recorded data.



Figure 1. Engine and radiator end of test article G-01 – a Cummins 4BTA3.9-powered 62.5 kVA generating set on its steel skid frame, prepared for pre-delivery load-bank testing.

2.2 Test article

Table 2. Principal data for the set used in the detailed load test (G-01).

Parameter	Value
Engine	Cummins 4BTA3.9-G2; 3.9 L, 4-cylinder, turbocharged and after-cooled; ≈ 60 kW _m (prime)
Alternator	Stamford-type UCI224, brushless, self-excited, 3-phase
Output	400/230 V, 3-phase, 4-wire, 50 Hz, 1500 rpm
Prime / standby rating	62.5 kVA / 50 kW at 0.8 pf (PRP); 68.75 kVA / 55 kW (ESP); rated current 90.2 A
Controller	Deep Sea DSE6020 – auto-start, instrumentation, shutdown protections
Cooling	Engine-mounted radiator, rated for 45 °C ambient

2.3 Recorded results

Table 3 gives the step-load data I recorded for set G-01. I calculated specific fuel consumption (SFC) and brake-thermal efficiency from the measured volumetric fuel rate, taking diesel density as 0.84 kg/L and lower heating value as 42.7 MJ/kg.

Table 3. Step-load test results for set G-01 (400 V, 3-phase, 50 Hz, 0.8 pf).

Load (%)	Load (kW)	Line I (A)	Volt. (V)	Freq. (Hz)	Oil P. (bar)	Coolant (°C)	Exhaust (°C)	Fuel (L/h)	SFC (g/kWh)
0	0.0	0	402	50.6	4.3	70	175	2.9	–
25	12.5	22.6	401	50.3	4.2	76	255	5.7	383
50	25.0	45.2	400	50.1	4.1	81	340	9.0	302
75	37.5	68.1	398	49.8	3.9	86	430	12.1	271
100	50.0	91.3	395	49.6	3.8	90	512	15.7	264
110	55.0	101.6	391	49.3	3.7	93	548	17.8	272

2.4 Discussion

Fuel consumption and specific fuel consumption. Fuel flow rises almost linearly with load, but SFC follows a U-shaped curve (Figure 2). It is very high at light load (383 g/kWh at 25%), falls to a minimum of 264 g/kWh at full load, and rises slightly again under overload. The high light-load figure reflects the fixed friction and pumping losses of the engine being spread over a small useful output.

Efficiency. Brake-thermal efficiency (Figure 3) mirrors the SFC curve: 22% at 25% load, rising to about 32% between 75% and 100% load. Operating the set below roughly 40% of its rating therefore wastes a large fraction of the fuel energy and, as discussed in Section 3, also harms the engine.

Voltage and frequency regulation. Figure 4 shows that frequency fell from 50.6 Hz at no load to 49.3 Hz at 110% load – a governor droop of about 2.6% and within the $\pm 1\%$ steady-state band expected of an electronically governed set. Line voltage stayed between 402 V and 391 V, inside $\pm 3\%$, confirming healthy AVR performance. Set G-01 met ISO 8528 performance class G3.

Thermal behaviour. Exhaust-gas temperature rose steadily to 548 °C at 110% load, comfortably below the 620 °C manufacturer limit, and coolant temperature peaked at 93 °C against a 104 °C alarm setting (Figure 5). Oil pressure stayed healthy (3.7–4.3 bar). No parameter approached a shutdown threshold during the test.

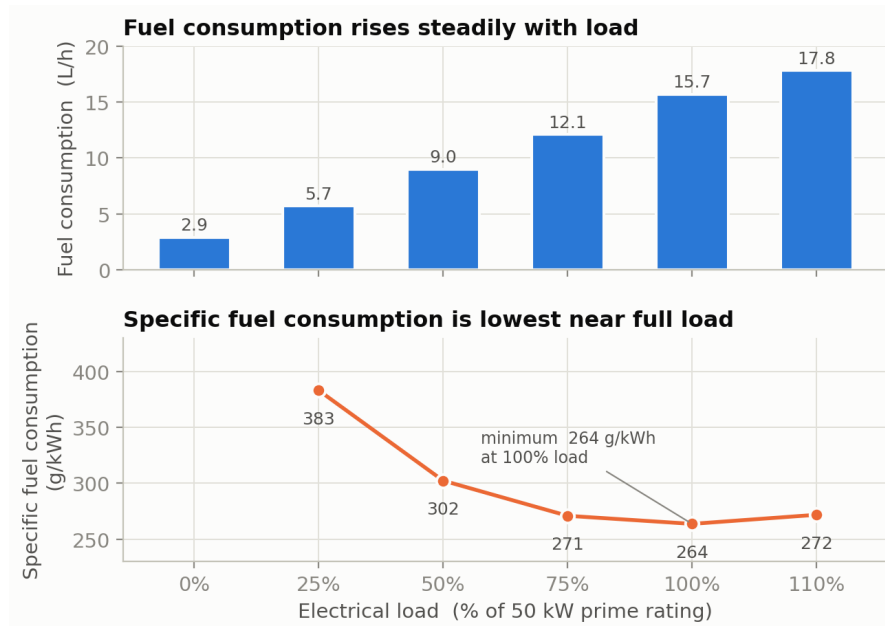


Figure 2. Fuel consumption (top) and specific fuel consumption (bottom) against electrical load for set G-01.

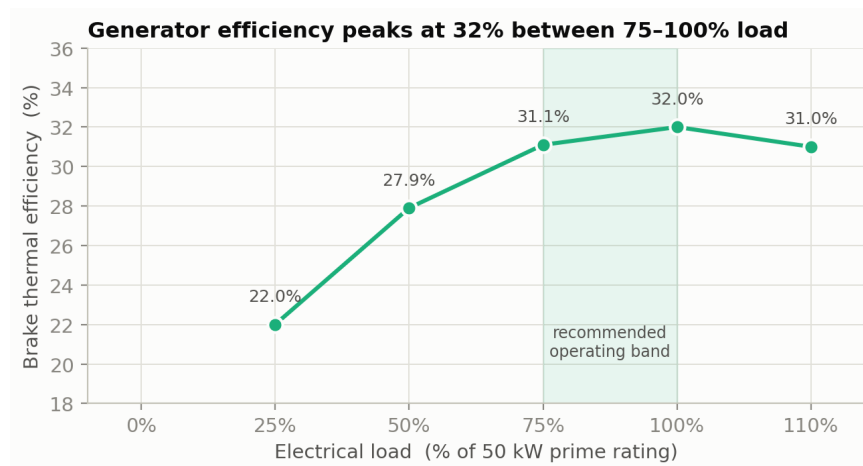


Figure 3. Brake-thermal efficiency against electrical load, with the recommended 75–100% operating band shaded.

2.5 Fleet test summary

I load-tested or assisted on three sets during my internship. Their results are summarised in Table 4.

Table 4. Summary of the three generating sets tested during the internship.

ID	Engine	Alternator	Rating (ESP)	Outcome
G-01	Cummins 4BTA3.9-G2	Stamford UCI224	68.75 kVA / 55 kW	Pass – all parameters within ISO 8528 class G3.
G-02	Cummins 6BTA5.9-G2	Stamford UCI274D	137.5 kVA / 110 kW	Pass after AVR re-tuning (regulation 4.8% → 2.1%).
G-03	Perkins 1104C-44TAG2	Leroy-Somer TAL-A	96 kVA / 77 kW	Conditional – thermostat replaced (coolant 101°C at 75%), re-tested Pass.

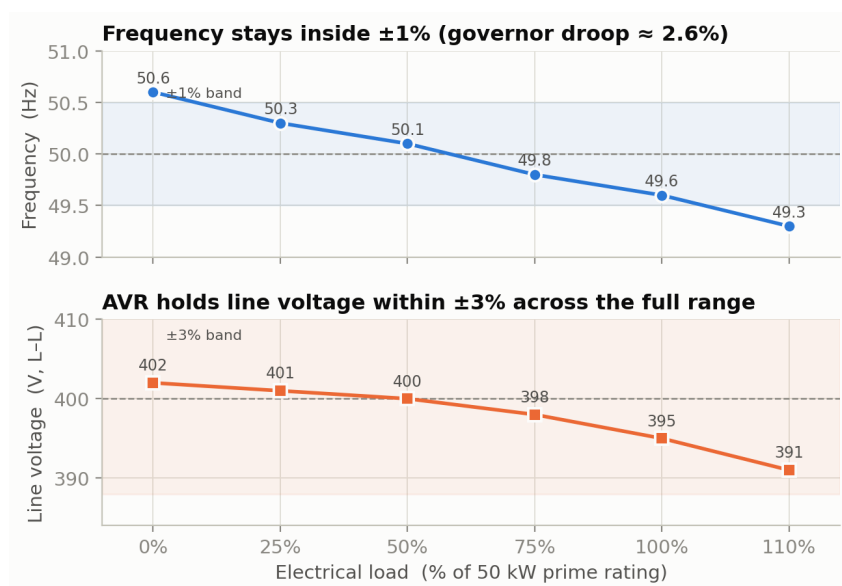


Figure 4. Frequency (top) and line voltage (bottom) against load, with the acceptance bands shaded.

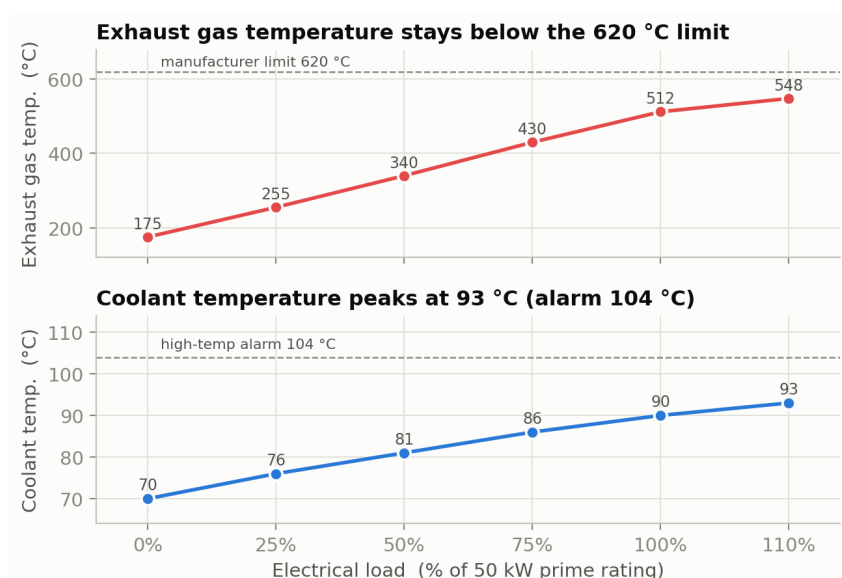


Figure 5. Exhaust-gas temperature (top) and coolant temperature (bottom) against load, with manufacturer limits shown.

3 Applied Research: Diesel Generators under Variable Load

Alongside the practical work I studied the topic “*Performance Analysis and Operational Optimization of Diesel Generator Sets under Variable Load Conditions.*” The main findings, and how I applied them at WMS, are summarised below.

- **The efficiency curve is strongly non-linear.** Brake-specific fuel consumption is lowest at roughly 70–90% load and deteriorates sharply below 30–40%. The G-01 test data in Section 2 are a direct example.
- **Sustained light-load running is damaging.** Below about 30% load a diesel engine runs cool, combustion is incomplete, and unburnt fuel and soot accumulate in the cylinders, exhaust and turbocharger – “wet stacking.” Over time this glazes the cylinder bores, fouls injectors and the turbo, and raises maintenance cost and smoke.
- **Mitigation strategies.** (i) right-size the set to the actual load; (ii) schedule periodic high-

load exercise runs or load-bank sessions to burn off deposits; (iii) use two or more smaller sets in parallel with load-dependent sequencing so each runs near its optimum; (iv) hybridise with solar plus battery storage to remove low-load hours; (v) consider variable-speed generating sets for highly variable loads.

- **Maintenance should be condition-based.** Oil-change and overhaul intervals are better set from operating hours, load profile and used-oil analysis than from the calendar alone, because a lightly loaded set degrades oil and accumulates deposits faster per running hour.
- **Transient response** on block loading depends on the engine's torque margin, the governor type and the alternator sizing; electronic governors and generously sized alternators give smaller voltage and frequency dips.

Application at WMS. From these findings I put three recommendations to my advisor: (1) advise customers in the handover documentation to keep sets in the 75–100% band and to avoid prolonged operation below 30%; (2) include a weekly 30-minute exercise run at $\geq 40\%$ load in every maintenance contract; and (3) offer a solar-hybrid backup option (Section 5) for sites whose base load is small relative to the installed set.

4 Routine Maintenance

Under supervision, I carried out scheduled preventive-maintenance (PM) tasks on workshop and customer sets. Table 5 lists the tasks I performed.

Table 5. Preventive-maintenance tasks carried out during the internship.

System	Task performed	Interval basis
Lubrication	Drained and refilled ≈ 14 L SAE 15W-40 API CI-4 oil; fitted new spin-on filter.	250 h / 6 months
Fuel system	Replaced primary and secondary filter elements; drained water separator.	250 h
Air intake	Inspected and cleaned pre-cleaner; checked restriction indicator.	Weekly / 500 h
Cooling system	Checked coolant level and 50/50 concentration; pressure-tested cap; cleaned radiator core.	Weekly / 2-yearly
Electrical / battery	Cleaned terminals; checked electrolyte; open-circuit voltage 12.9 V; load test passed.	Monthly
Belts & hoses	Checked tension and condition; no cracking or chafing found.	Weekly
Controls & protection	Tested auto-start, low-oil-pressure and high-temperature shutdowns, and emergency stop.	Monthly
Exercise run	30-minute run at $\approx 40\%$ load with parameters logged.	Weekly

Faults I found and helped correct. The late-opening thermostat on set G-03 (Section 2.5); out-of-tolerance voltage regulation on G-02, corrected by adjusting the AVR stability setting; a weeping fuel-return banjo bolt on G-01, cured with new copper washers; and a tripped air-filter restriction indicator on a stored 6-cylinder set, cleared by replacing the element.

5 Standalone Solar Integration and Backup Solutions

5.1 System studied

In my final week I worked with the solar team on an 11.8 kWp three-phase hybrid system at a commercial customer in Lahore, and then evaluated the same architecture as a backup-power option for the WMS workshop itself. On site I terminated the DC strings at the inverter and inspected the module clamps, rails and earthing on the ground-mounted array (Figure 6). System data are given in Table 6.



Figure 6. Left: termination of the DC strings and protection devices at the WOXEN three-phase hybrid inverter. Right: inspection of module clamps and rail earthing on the ground-mounted array.

Table 6. Specification of the 11.8 kWp hybrid system studied.

Item	Specification
PV modules	20 × 590 W bifacial mono-PERC (Tier-1), 11.8 kWp total
Array configuration	2 strings × 10 modules; string open-circuit voltage ≈ 512 V
Inverter	WOXEN 12 kW three-phase hybrid string inverter; 2 MPPT inputs; 98.2% peak efficiency
Battery	15 kWh LiFePO ₄ , 51.2 V, $\approx 90\%$ usable depth of discharge, >6000 cycles
Mounting	Ground-mounted hot-dip-galvanised A-frame; 25° tilt; azimuth 180° (due south)
Protection	DC Type-2 surge arrester, string fuses/MCBs, AC surge arrester, 40 A MCB and RCD; dedicated earth electrode $\leq 5 \Omega$
Metering	Bi-directional net meter; inverter Wi-Fi monitoring

5.2 Estimated energy yield for Lahore

I estimated monthly generation from long-term peak-sun-hour (PSH) data for Lahore and a performance ratio of 0.78 (which accounts for temperature, soiling, wiring, inverter and mismatch losses). My results are given in Table 7 and Figure 7.

Table 7. Modelled monthly generation, 11.8 kWp array, Lahore (PR = 0.78).

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Year
PSH (kWh/m ² /day)	3.3	4.1	5.0	5.9	6.4	6.1	5.3	5.1	5.2	4.7	3.7	3.1	4.7
Generation (kWh)	942	1057	1427	1629	1826	1684	1512	1452	1436	1341	1022	884	16,212

5.3 Diesel-only versus solar-hybrid backup

The WMS workshop experiences roughly five hours of utility outage per day and a critical backup demand averaging 6.4 kW, i.e. about 32 kWh/day. I compared two ways of supplying this (Table 8, Figure 8): running the existing diesel set alone, or a solar-hybrid system in which the PV array and battery serve most of the energy and the diesel set starts only as a final backup.

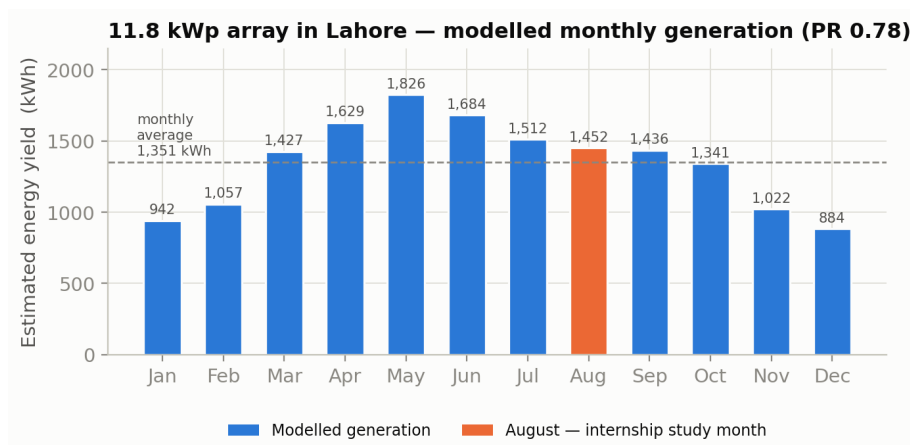


Figure 7. Modelled monthly generation for the 11.8 kWp array in Lahore. Annual total $\approx 16,200$ kWh; specific yield $\approx 1,374$ kWh/kWp.

Table 8. Backup-supply comparison for a 32 kWh/day requirement (diesel at PKR 275/L; CO₂ factor 2.68 kg/L).

Metric	Diesel set only	Solar-hybrid + set backup
Energy served by diesel	32 kWh/day	≈ 5 kWh/day
Diesel consumed	≈ 10.2 L/day	≈ 1.7 L/day
Fuel cost	$\approx$ PKR 2,820/day	$\approx$ PKR 470/day
Monthly fuel cost	$\approx$ PKR 84,500	$\approx$ PKR 14,000
Monthly maintenance	$\approx$ PKR 9,000	$\approx$ PKR 4,500
Monthly operating cost	$\approx$ PKR 93,500	$\approx$ PKR 18,500
Operational CO ₂	≈ 823 kg/month	≈ 137 kg/month
Indicative system capital cost	–	$\approx$ PKR 2.35 million
Simple payback vs. diesel-only	–	≈ 31 months

Workshop backup supply: diesel-only vs. solar-hybrid (32 kWh/day need)

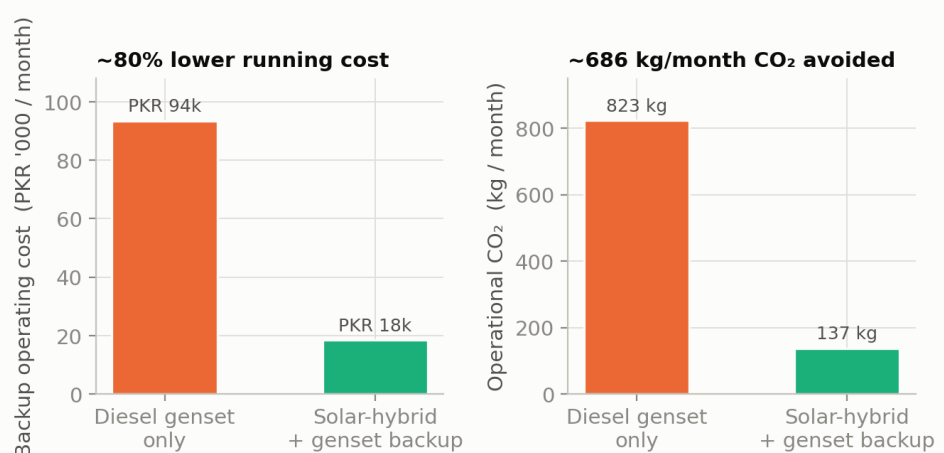


Figure 8. Monthly backup operating cost and operational CO₂ for the two options.

5.4 Integration considerations

- **Control.** In hybrid-backup mode the solar inverter forms the supply during an outage and the diesel set auto-starts only on low battery state-of-charge or high demand. The inverter must support a generator input and AC-coupled charging (the WOXEN unit does).
- **Keeping the set in its efficient band.** When the diesel set does run, the inverter should charge the battery so that the set is loaded to at least 40–50%, avoiding the light-load

problems described in Section 4.

- **Load management.** Non-critical loads (air-conditioning, non-urgent machining) are shed during outages so the battery lasts longer.
- **Compliance.** Anti-islanding protection and net-metering approval (LESCO / AEDB) are required for the grid-connected portion.
- **Operation and maintenance.** Modules need cleaning every 2–3 weeks in Lahore’s dust; an annual thermographic scan and a battery/BMS check are recommended.

6 Learning Outcomes

During the internship I gained:

- practical experience of generating-set pre-delivery inspection, resistive load-bank testing, instrumentation and ISO 8528 acceptance criteria;
- an understanding of diesel-engine performance – specific fuel consumption, efficiency, wet stacking and load management – plus hands-on preventive maintenance and fault diagnosis;
- the fundamentals of solar PV system design: string sizing, inverter and battery selection, protection and earthing, and energy-yield estimation;
- practice in techno-economic comparison of energy options, technical report writing, workshop health and safety, and teamwork with technicians.

7 Conclusions and Recommendations

- My detailed load test showed that set G-01 meets its ratings and ISO 8528 class G3, with best efficiency (32%) and lowest specific fuel consumption (264 g/kWh) between 75% and 100% load. All three sets I worked on were made fit for dispatch, two of them after minor corrective work.
- WMS should adopt a standard written load-test datasheet – of the kind I developed during this internship – for every set, and should include operating guidance (75–100% band, weekly exercise run) in its handover documentation.
- For sites with a small base load, a solar-hybrid backup cuts the workshop’s standby running cost by about 80% and its operational CO₂ by about 83%, with a payback under three years; it is recommended both as a customer offering and for the workshop itself.

References

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